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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction To Internet Of Things (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

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Week 6 ()

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Week 11 ()

- Lecture 51: Smart Grid- I (unit?unit=111&lesson=112)
- Lecture 52: Smart Grid- II (unit?unit=111&lesson=113)

Lecture 53: Industrial Internet of Things- I (unit?unit=111&lesson=114)
Lecture 54: Industrial Internet of Things- II (unit?unit=111&lesson=115)
Lecture 55: Data Handling and Analytics- I (unit?unit=111&lesson=116)
Lecture material of Week 11 (unit?unit=111&lesson=117)
Week 11 Feedback Form (unit?unit=111&lesson=314)
Quiz: Week 11 : Assignment 11 (assessment?name=374)
Week 12 ()
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Week 11 : Assignment 11

Your last recorded submission was on 2026-04-07, 14:26 IST **Due date: 2026-04-08, 23:59 IST.**

1) Which of the following is also used as a name for Smart Grids? *1 point*

a. Energy internet

b. Electricity with a hand

c. Electrolyte

d. All of these

- (●) a.
- ( ) b.
- ( ) c.
- ( ) d.

2) Smart Grid enables which type of energy flow? *1 point*

a. Bidirectional

b. Unidirectional

c. Both a and b

d. Neither a nor b

- (●) a.
- ( ) b.
- ( ) c.
- ( ) d.

3) *1 point*

Fill in the blank.
In _____, a home can draw power from distributed resources.

a. Peaking

b. Islanding

c. Neither a nor b

d. Both a and b

- ( ) a.
- (●) b.
- ( ) c.
- ( ) d.

4) *1 point*

Fill in the blank.
_____ samples voltage and current with a fixed sample rate at the installed location.

a. PEVs

b. DAUs

c. PMUs

d. None of these

- ( ) a.
- ( ) b.
- (●) c.

○ d.

5) *1 point*

Fill in the blank.

_____ is carried out by compromising the availability of a grid component.

a. Gateway

b. Data injection attacks

c. AMH

d. Denial of service

○ a.
○ b.
○ c.
◉ d.

6) *1 point*

Which of the following are types of Dynamic System Attacks?

a. Replay attacks

b. Dynamic data injection attacks (D-DIA)

c. Covert attacks

d. All of these

○ a.
○ b.
○ c.
◉ d.

7) *1 point*

State True or False.

Industrial Internet of Things (IIoT) follows the "rip & replace" approach.

a. True

b. False

○ a.
◉ b.

8) *1 point*

State True or False.
Mechanized production is a part of the 4th Industrial Revolution.

a. True

b. False

- ○ a.
- ◉ b.

9) ***1 point***

Match the Industrial Revolution with the corresponding milestone:
1) 1st 2) 2nd 3) 3rd 4) 4th

a. 1-Mechanized production, 2-Mass production, 3-Internet evolution & automation, 4-Industrial Internet of Things

b. 1-Mass production, 2-Mechanized production, 3-IIoT, 4-Internet evolution & automation

c. 1-Internet evolution & automation, 2-Mass production, 3-Mechanized production, 4-IIoT

d. 1-IIoT, 2-Internet evolution & automation, 3-Mass production, 4-Mechanized production

- ◉ a.
- ○ b.
- ○ c.
- ○ d.

10) ***1 point***

Which of the following is used to improve interoperability across different systems?

a. Standardization

b. Optimization

c. Monitoring

d. None of these

- ◉ a.
- ○ b.
- ○ c.
- ○ d.

11) ***1 point***

Fill in the blank.
_____ is a primary challenge in IIoT.

a. Worker health and safety

b. Optimised operations

c. Regulatory compliance

d. Integrating existing infrastructure into new IIoT infrastructure

- ○ a.
- ○ b.
- ○ c.
- ◉ d.

12) *1 point*

State True or False.
Lack of vision and leadership is not a hindrance in the path of success of IIoT.

a. True

b. False

- ○ a.
- ◉ b.

13) *1 point*

State True or False.
Big data is characterized by 7 Vs.

a. True

b. False

- ◉ a.
- ○ b.

14) *1 point*

What is the correct flow of data processing?

a. Acquisition > Generation > Storage > Analysis

b. Generation > Storage > Analysis > Acquisition

c. Generation > Acquisition > Storage > Analysis

d. None of these

- ○ a.
- ○ b.
- ◉ c.
- ○ d.

15) Hadoop is primarily used for: ***1 point***

a. Distributed processing of large datasets

b. Large clusters of computers

c. All of these

d. Neither a nor b

- ○ a.
- ○ b.
- ◉ c.
- ○ d.

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers